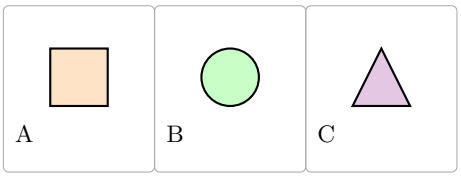
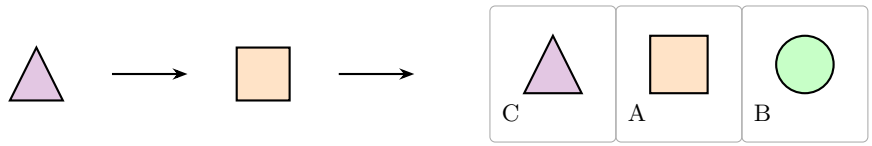
Problem 1: Make every different row using these three cards once each.



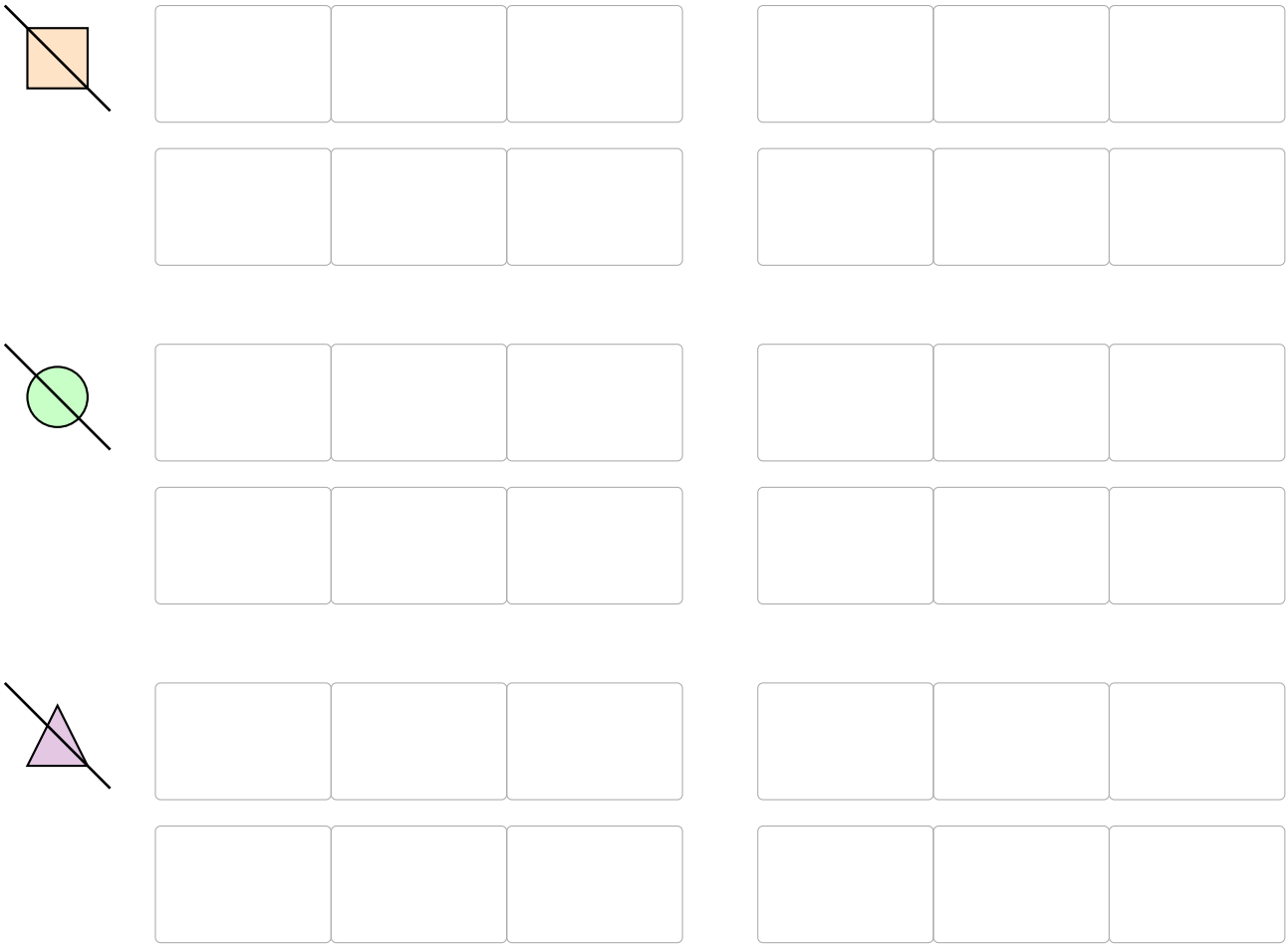
1	2	3

Mix the three picture cards in a cup. Each card has the same chance. Draw the first without looking and keep it out. Draw the second from the two left. Put the last card at the end. Return all cards for a new row.

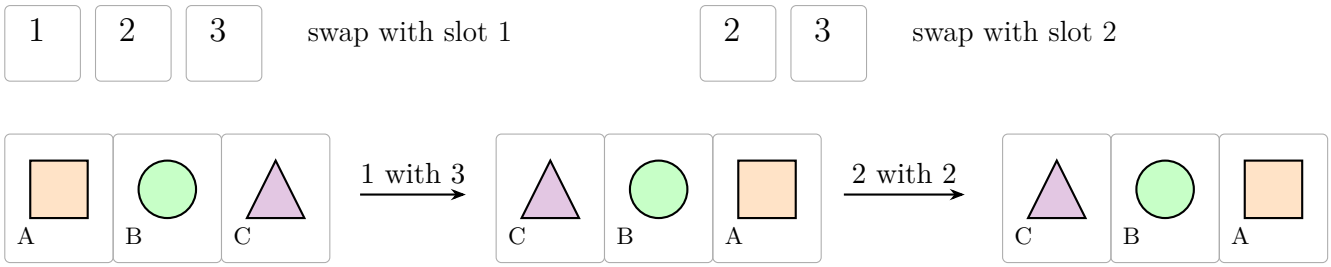


Example: choose triangle, then square; circle is left.

Problem 2: For each crossed-out first card, find every row still possible. Could two different first-two draws make the same row?



Start A B C. Mix tickets 1, 2, 3 in the cup; draw and swap its slot with slot 1. Empty the cup, then mix tickets 2, 3; draw and swap its slot with slot 2. At each draw, all tickets in the cup have equal chances. Reset both sets for every new row. A slot may swap with itself.



Problem 3: Start with A B C and choose tickets to make every row. Can any row be made by two different ticket stories?

